

1) Velocity of the water jet relative to the deflector:

$$V_r = 30 - 10$$

$$= 20 \text{ ms}^{-1}$$

$$\sum F = \dot{m} (V_{2x} - V_{1x})$$

$$F + mg = \dot{m} (20 - 20 \cos 30^\circ)$$

$$F + 0.1 \times 9.81 = 1.23 (20) \left(\frac{10 \times 10^{-2}}{2} \right)^2 \times (20 - 20 \cos 30^\circ)$$

$$F = -0.4633008631$$

$$\approx -0.463 \text{ N}$$

$$\begin{aligned}
 2a) \quad \dot{m} &= \rho V A \\
 &= 160(15)(20 \times 10^{-2} \times 4) \\
 &= 1920 \text{ kg s}^{-1}
 \end{aligned}$$

$$\cancel{\frac{p}{\rho g}} + \frac{V_1^2}{2g} + \cancel{z_1} = \cancel{\frac{p}{\rho g}} + \frac{V_2^2}{2g} + \cancel{z_2}$$

$$\frac{15^2}{2g} = \frac{V_2^2}{2g}$$

$$\therefore V_2 = 15 \text{ ms}^{-1}$$

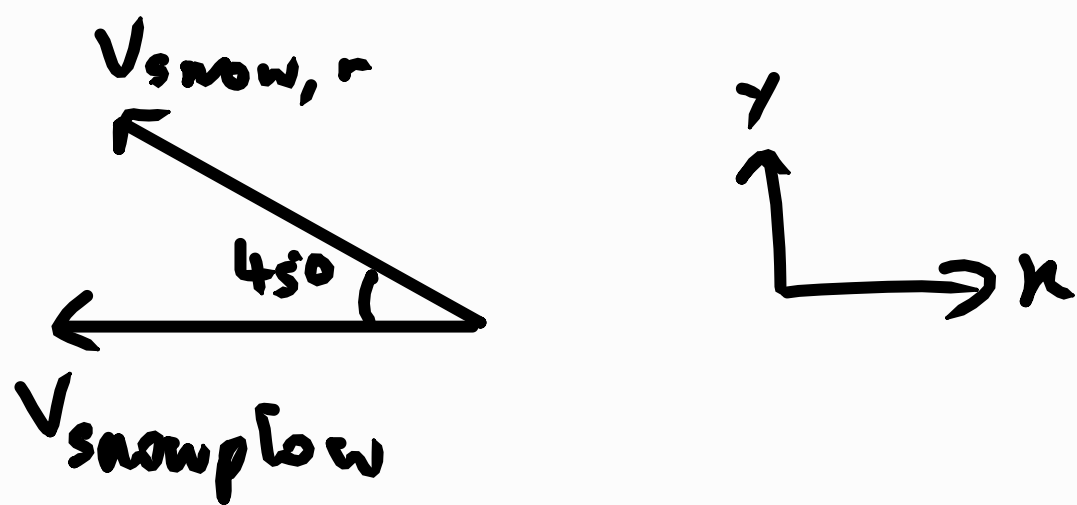
$$\Sigma F = \dot{m}(V_{2x} - V_{1x})$$

$$-F = 1920(-15 \cos 45^\circ - 15)$$

$$F = 49.1646753 \times 10^3$$

$$\approx 49.16 \times 10^3 \text{ N}$$

2b) The relative velocity of the snow is 15 ms^{-1} with respect to the moving snowplow, at 15 ms^{-1} .



Absolute velocity of the snow:

$$V_{\text{snow}, x} = -V_{\text{snow}, r} \cos 45^\circ - V_{\text{snowplow}}$$

$$= -15 \cos 45^\circ - 15$$

$$= -\frac{30 + 15\sqrt{2}}{2}$$

$$\approx -25.61 \text{ ms}^{-1}$$

$$V_{\text{snow}, y} = V_{\text{snow}, r} \sin 45^\circ$$

$$= 15 \sin 45^\circ$$

$$= \frac{15\sqrt{2}}{2} \approx 10.61 \text{ ms}^{-1}$$

$$V_{\text{snow}, \text{abs}} = \sqrt{\left(-\frac{30 + 15\sqrt{2}}{2}\right)^2 + \left(\frac{15\sqrt{2}}{2}\right)^2}$$

$$= 27.71638598 \text{ ms}^{-1}$$

$$\approx 27.7 \text{ ms}^{-1}$$

$$\theta = \arctan\left(\frac{10.61}{25.61}\right) = 22.5^\circ$$

$$\begin{aligned}
 3) \quad Q &= 60 \text{ L min}^{-1} \\
 &= \frac{60 \times 10^{-3}}{60} \\
 &= 1 \times 10^{-3} \text{ m}^3 \text{ s}^{-1}
 \end{aligned}$$

$$Q = V_2 2A_2$$

$$\begin{aligned}
 V_2 &= \frac{Q}{2A_2} = \frac{1 \times 10^{-3}}{2 \times 0.26 \times 10^{-4}} \\
 &= \frac{250}{13} \text{ m s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 \dot{m} &= 1000 \times 1 \times 10^{-3} \\
 &= 1 \text{ kg s}^{-1}
 \end{aligned}$$

For the sprinkler:

$$\begin{aligned}
 T &= \dot{m} r (V_2 - \omega r) \\
 &= 1(0.2) \left(\frac{250}{13} - 0.2\omega \right) \\
 &= \frac{50}{13} - 0.04\omega
 \end{aligned}$$

a) When $\omega = 0$,

$$T = \frac{50}{13} \approx 3.85 \text{ Nm}$$

b) When $\omega = 500 \text{ rpm}$,

$$\begin{aligned}
 T &= \frac{50}{13} - 0.04 \left(\frac{500(2\pi)}{60} \right) \\
 &= 1.751758744 \approx 1.75 \text{ Nm}
 \end{aligned}$$

c) When $T = 0$,

$$0 = \frac{50}{13} - 0.04\omega \rightarrow \omega = \frac{1250}{13} \approx 96.15 \text{ rad s}^{-1}$$

$$\begin{aligned}
 4) \quad Q &= 5 \text{ L s}^{-1} \\
 &= 5 \times 10^{-3} \text{ m}^3 \text{ s}^{-1} \\
 \dot{m} &= \rho Q = 1000 \times 5 \times 10^{-3} \\
 &= 5 \text{ kg s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 Q &= V_2 (3A_2) \\
 V_2 &= \frac{5 \times 10^{-3}}{3(18 \times 10^{-6})} \\
 &= \frac{2500}{27}
 \end{aligned}$$

$$\begin{aligned}
 T &= \dot{m} r (V \cos \theta - r\omega) \\
 &= 5(0.5) \left(\frac{2500}{27} \cos \theta - 0.5\omega \right) \\
 &= 2.5 \left(\frac{2500}{27} \cos \theta - 0.5\omega \right)
 \end{aligned}$$

a) When $\theta = 0$, $\omega = 0$,

$$\begin{aligned}
 T &= 2.5 \left(\frac{2500}{27} \cos 0^\circ - 0.5(0) \right) \\
 &= \frac{6250}{27} \\
 &\approx 231 \text{ Nm}
 \end{aligned}$$

b) When $T = 0$, $\theta = 30^\circ$,

$$\begin{aligned}
 0 &= 2.5 \left(\frac{2500}{27} \cos 30^\circ - 0.5\omega \right) \\
 \omega &= 160.3750748 \\
 &\approx 160 \text{ rad s}^{-1}
 \end{aligned}$$